

## ABDELRAHMAN GAMAL ZAYED

### AI / Machine Learning Intern | AI/ML Engineering Student

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drive.google.com/file/d/1HPSAFTZxcPxehcgTdr11Tuwy\_9khec-P/view

## SUMMARY

Motivated Artificial Intelligence Engineering student targeting an AI / Machine Learning Intern role, with hands-on project development in supervised learning, data-driven solutions, intelligent solutions, computer vision, NLP fundamentals, and under-review research in solar power forecasting. Skilled in data collection, data cleaning, exploratory data analysis, feature extraction, feature engineering, model evaluation, validation metrics, data visualization, and technical documentation. Strong collaboration, analytical thinking, and motivation to support AI, Machine Learning, Data Science, Computer Science, and related fields.

## EDUCATION

### Bachelor of Engineering in Artificial Intelligence Engineering | New Ismailia National University

Oct 2023 - Jul 2028 | Ismailia, Egypt

• GPA: 3.65 / 4.00

• Relevant coursework: Machine Learning, Deep Learning, Data Structures, OOP, Statistics, Algorithms, Data Analysis, Artificial Intelligence, Computer Science, Data Science fundamentals.

## TECHNICAL SKILLS

Programming: Python, C, C++, Object-Oriented Programming, Data Structures, Algorithms

AI / ML: Machine Learning, Supervised Learning, Deep Learning, Neural Networks, Computer Vision, Natural Language Processing, NLP, Text Classification, Image Classification, Transfer Learning

Data: Data Collection, Data Cleaning, Data Preprocessing, Exploratory Data Analysis, Feature Extraction, Feature Engineering, Feature Selection, Data Visualization

Modeling: Logistic Regression, Random Forest, SVM, Decision Tree, Cross-Validation, Hyperparameter Tuning, Model Deployment Basics, Model Evaluation, Validation Metrics

Libraries & Tools: Scikit-learn, Pandas, NumPy, TensorFlow, Keras, PyTorch basics, OpenCV basics, Matplotlib, Seaborn, Jupyter Notebook, Git, PyCharm

Metrics: Accuracy, Precision, Recall, F1-score, Confusion Matrix, ROC-AUC | Soft Skills: Motivation, Collaboration, Analytical Thinking, Problem Solving, Teamwork | Languages: Arabic Native, English Intermediate

## EXPERIENCE

Deep Learning Trainee | Egyptian Engineers Syndicate

Sep 2025 | Online | 2-week training

• Completed focused training in neural networks, training and validation logic, overfitting control, transfer learning fundamentals, and practical deep learning applications.

Machine Learning Intern | National Telecommunication Institute (NTI)

Jul 2025 | Remote

• Built a diabetes prediction workflow for structured health data, covering data cleaning, preprocessing, feature handling, model training, validation metrics, evaluation, and technical findings documentation.  
• Benchmarked 4 supervised learning algorithms - Logistic Regression, Random Forest, SVM, and Decision Tree - using 5 metrics: accuracy, precision, recall, F1-score, and confusion matrix.

AI & Machine Learning Trainee | SprintUp Program by Sprints

Apr 2025 - Jun 2025 | Remote

• Developed an end-to-end machine learning workflow using Python, Scikit-learn, exploratory data analysis, data visualization, feature engineering, model comparison, and result interpretation.  
• Applied analytical thinking, cross-validation, and hyperparameter tuning concepts to support model selection, reduce overfitting risk, and communicate technical findings.

## RESEARCH

### Research Paper - Under Review | Hybrid Metaheuristic Model for Solar Power Forecasting

#### ELM | Improved Harris Hawks Optimization (IHO) | XGBoost | CatBoost | Transformer | Time-Series Forecasting

• Co-authored an under-review research paper integrating Extreme Learning Machine (ELM) with Improved Harris Hawks Optimization (IHO) for solar power forecasting using PV installation data.  
• Evaluated the model against CatBoost, XGBoost, Transformer, and standalone ELM baselines; reported  $R^2 = 0.993$ ,  $RMSE = 0.167$ , and  $MAPE = 3.75\%$  on the spring dataset.

## PROJECTS

### Diabetes Prediction Classification Model | Machine Learning Project

Python | Scikit-learn | Pandas | NumPy | Matplotlib | Seaborn

• Built a classification pipeline for diabetes risk prediction using structured tabular health data, data cleaning, feature engineering, supervised learning, and imbalance-aware evaluation.  
• Compared 4 ML models and documented technical findings using accuracy, precision, recall, F1-score, ROC-AUC, confusion matrices, and data visualization for data-driven model selection.

### Facial Emotion Recognition Model | Computer Vision / Deep Learning Project

TensorFlow | Keras | OpenCV basics | Transfer Learning | Image Classification

• Developed a 7-class facial emotion recognition workflow using image preprocessing, augmentation, CNN/transfer learning concepts, validation monitoring, and class-level error analysis.  
• Used confusion matrices, validation curves, and analytical thinking to identify overfitting patterns and improve evaluation quality for intelligent computer vision solutions.

## LEADERSHIP & VOLUNTEERING

### AI Head | Microsoft Students Club - Suez Canal University (SCU)

Student Activity

• Lead AI peer learning and support collaboration in machine learning, deep learning, NLP, computer vision, and project development.

### Volunteer | Egyptian Youth Entity (EYE)

Community & Youth Development

• Contribute to youth development initiatives through teamwork, event organization, volunteer engagement, and collaboration.